

ture. This meant that each bus component had its own path. "You could use a pattern or state trigger to find an event of interest and then just visually decode the data on the bus with such an architecture," explains Bhansali. "Modern embedded designs, however, use serial architecture, meaning the bus data is sent serially. Serial architecture is used because less board space is required, the cost is less, embedded clocks can be used and power requirement is less."

PC interconnects

Development of PC interconnect technologies has been the key driver for PC-based instrumentation. "PC-based digital oscilloscopes are growing in popularity as these offer a considerable cost saving over their bench-top equivalent," informs Bhansali. "The reason for cost saving is obvious—by using a PC already sitting on your desk, you can leverage a large colour display, fast processor, disk drives and keyboard effectively for free," he adds.

PC-based scopes are available in two different forms: internal and external. Internal PC-based scopes are usually in the form of PCI-format plug-in cards. Bhansali says that the biggest disadvantage of PC cards is the noise; the inside of a personal computer can be a very noisy electrical environment and some cards suffer from it. Another disadvantage is portability; PC cards are tied to use with one desktop PC.

External PC-based oscilloscopes take the form of a small box that connects to the PC via USB or parallel port. As all the analogue electronics is kept outside the PC, the issue of noise is resolved. These are portable and can be used with either desktop or laptop PCs.

A personal computer has many features in terms of hardware and software. PC-based scopes leverage all these to provide a large display, easy storage and sharing of waveforms, easy analysis, export of data samples to Excel and WordPress, high-speed USB connection and compact form factor. Apart from that, any new functionality can be implemented with software upgrade.

PC-based scopes can also offer high

resolutions of 10, 12 and 14 bits, which are used for debugging high-fidelity applications like audio.

Displays for enhanced view

Oscilloscopes are meant to view signals, and if the viewing is not good, their primary work is not done! Scopes today have full-range colour displays, LCD and plasma displays with screen size ranging from 14 cm to 39 cm. Modular oscilloscopes without their own display screens, can even be hooked up to giant digital signage in shopping malls. High-speed sampling and high resolution compliment the display of waveforms.

High-speed embedded designs require an oscilloscope to provide integrated high-bandwidth digital channels with analogue channels

With advanced software tools, you can also view a 3-D graphical representation of the signal in time and frequency, and convolution and histograms, giving you a better understanding of the overall signal spectrum.

"The early digital scopes did a great job of data capture. However, these lacked the unique persistence characteristics of analogue scopes' phosphor display that showed how the signal changed over time," shares Bhansali. He adds, "Introduction of high-definition XGA display and digital phosphor displays eliminated the last remaining advantage of analogue scopes. With a digital phosphor display, waveform after waveform is overlaid on the display, making the display more intense." DSOs are continually growing as, like analogue real-time scopes, these allow you to view the glitches and rare spots.

The display plays an important role when it comes to simultaneous display of multiple channels and measurement of parameters. "When you want to view, say, four channel traces, a single grid screen is split into four units, one for each trace. However, you do not get full ADC resolution on a single-grid display as multiple attenuated waveforms are placed one above the other. So there

is a trade-off between waveform display and parameter measurement. The introduction of multi-grid display utilises the full dynamic range of the ADC for each trace and helps to get more accurate measurements," explains Rajendran.

New features in scopes

While scopes are still used for viewing waveform shape and measuring different parameters, it is becoming increasingly common for people to use them as an automated verification tool. These are used to make sure that the device meets the specification requirements of serial data standards. They also include

several application protocols and compliance analysis, vector signal analysis, jitter analysis, noise figure analysis and complex triggering.

Compliance and protocol test. "Various application-specific compliance test software can be loaded on the oscilloscopes to cater to the needs of different market segments. Test software like certified wireless USB, FireWire IEEE 1394 a/b and USB 2 are used in computer electronics, whereas digital video interface (DVI) and high-definition multimedia interface (HDMI) find their use in media. Controller area network (CAN), LAN and FlexRay are used in the automotive industry," shares Bhatia.

"Oscilloscopes are used to debug complex protocols through accurate and complex triggering mechanisms wherein the user gets the power to correlate protocol decode with the physical layer signal."

Serial data equalisation and de-embedding. "Apart from the compliance testing for standards, there is also a strong need to embed or de-embed a fixture, channel or be able to apply an FFE/DFE equalisation on the data in order to analyse the complete serial data link," shares Dhruv.

Equalisation techniques used in dig-